Yuping Zheng

Department of Computer Science
University of Virginia

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Education

- **University of Virginia** — Charlotteville, VA
  Ph.D. student, Computer Science — *Aug. 2026 – Present*
  Rotation Advisor: Prof. Xi Peng
- **The Chinese University of Hong Kong (Shenzhen)** — Shenzhen, China
  Bachelor of Science, Data Science and Big Data Technology — *Aug. 2022 – June 2026*

Research Interests

I am interested in how AI systems learn, represent information, and reason. I aim to develop and test falsifiable hypotheses about these processes, drawing on ideas from cognitive science, neuroscience, mathematics, and related fields.

Publications and Preprints (Google Scholar)

1. Joshua Monickaraj, Hasin Rehana, Ani Bernardi, **Yuping Zheng**, Taiyu Lin, Le Liu, Amogh Madireddi, Leo Yeh, Jie Zheng, Junguk Hur, and Yongqun He.
   Vaxjo 2.0: A Web-based Vaccine Adjuvant Knowledgebase Powered by Ontology and Large Language Models. Frontiers in Cellular and Infection Microbiology, 2026. **paper**
2. **Yuping Zheng** and Yongqun He.
   VaxjoGNN: A Graph Neural Network for Ontology-Grounded Vaccine Adjuvant Recommendation. *bioRxiv* 2025. **paper**
3. Guorui Zheng, Xidong Wang, Juhao Liang, Nuo Chen, **Yuping Zheng**, Benyou Wang.
   Efficiently Democratizing Medical LLMs for 50 Languages via a Mixture of Language Family Experts. ICLR 2025. **paper**
4. Xingxian Li, **Yuping Zheng**, Joy Hu, Jie Zheng, Zhigang Wang, and Yongqun He.
   VaxLLM: Leveraging a Fine-tuned Large Language Model for Automated Vaccine Annotation. *bioRxiv* 2024. **paper**

Research Experience

DeepREALlab, UVA CS — Fall 2026 – Present

World Models for Longitudinal Medical Data

- Formulated a framework to jointly forecast tumor and liver-function changes from longitudinal imaging and multimodal clinical data for survival prediction in hepatocellular carcinoma (HCC).
- Conducted preliminary experiments on public datasets, observing improved performance in predicting Overall Survival relative to baseline methods.

Genlab, CUHK-Shenzhen — Feb. 2025 – May 2026

Undergraduate Researcher; Advisor: Zhen Liu

Representation Collapse in Discrete Flow Matching

- Designed learned token embeddings and neural schedule variants for discrete flow matching; built a reproducible training and evaluation pipeline on CIFAR-10.
- Observed near-one-dimensional collapse of 16-dimensional embeddings despite decreasing training loss; analyzed the collapse using singular-value spectra, cosine similarity, and PCA.
- Tested timestep reweighting toward high-noise regimes and an embedding self-reconstruction penalty, observing partial recovery of embedding rank.

He Group, University of Michigan Medical School — Feb. 2024 – May 2026

Undergraduate Researcher; Advisor: Yongqun He

VaxjoGNN: Ontology-Grounded Vaccine Adjuvant Recommendation

- Led development of a heterogeneous graph neural network for disease–adjuvant recommendation, incorporating pathway priors from the Vaccine Ontology to support generalization to unseen diseases.
- Achieved better results on held-out diseases, compared with text-similarity baselines.
- Conducted reliability analysis and identified overconfidence despite improved ranking performance, highlighting a gap between ranking quality and confidence calibration.

Vaxjo 2.0: A Web-based Vaccine Adjuvant Knowledgebase

- Developed a two-phase pipeline to convert PubMed articles into structured vaccine metadata for Vaxjo 2.0 data curation, and analyzed the resulting dataset.

CUHK-Shenzhen NLP Group — Aug. 2024 – Apr. 2025

Undergraduate Researcher; Advisor: Benyou Wang

Multilingual Medical LLMs via Mixture of Language Family Experts

- Analyzed cross-lingual information flow and evaluated final-layer-only training while preserving performance across 50+ languages; contributed routing experiments for post-MoE design.
- Built data pipelines for 2+ TB of multilingual medical text across 12 languages and implemented large-scale routing benchmarks in PyTorch.

Honors & Awards

- **Bowen Scholarship**, CUHK-Shenzhen — 2022 – 2026
  Merit-based scholarship covering 30% of undergrad tuition(approximately $20000 in total).
- **Dean's List**, CUHK-Shenzhen — 2024 – 2025